SpectralForge Studio: A Physically Comprehensive Multi-Excitation Fluorescence Hyperspectral Simulator and Experiment-Design Toolkit for Layered Tissue

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***Abstract*—Multi-excitation fluorescence hyperspectral imaging (ME-HSI) records an excitation–emission matrix at every pixel, which makes it attractive for label-free imaging of tissue, but the signal of a buried fluorophore is distorted by the scattering and absorbing layers above it, and real acquisitions carry no ground truth. We present SpectralForge Studio, an open simulator and experiment-design environment for ME-HSI of layered tissue. It combines a 3D scene editor, a physically comprehensive forward model (ray optics, dye photophysics, a layered-block three-dimensional Monte Carlo with a peel-off estimator and voxel inclusions, a literature tissue-optics library, and a camera chain with tunable-filter bands and Poisson–Gaussian noise), and a research toolkit: a study engine with provenance, Sobol' and Morris sensitivity analysis, linear unmixing, surrogate-based maximum-likelihood retrieval with profile-likelihood intervals, Fisher-information and Cramér–Rao analysis with greedy D-optimal channel selection, and a calibration module that turns the simulator into a digital twin of a laboratory instrument. The transport engine reproduces the MCML benchmark to better than $0.1\,\%$, conserves energy to 10^{-9}, and recovers the configured detector gain and read noise to $0.2\,\%$ and $0.1\,\%$. A calibrated twin predicts held-out excitation lines of measured collagen spectra with a mean spectral angle of 8.4° against 14.2° for the nominal instrument. In simulation studies on collagen-covered tissue we find [TODO: headline numbers from E2–E9: effective attenuation, share of redox-ratio variance caused by the cover, number of channels reaching the CRLB of the full protocol, abundance bias of linear unmixing]. The software is released as open source.**

***Index Terms*—Fluorescence hyperspectral imaging, Monte Carlo light transport, tissue optics, simulation, experiment design, Fisher information, sensitivity analysis, digital twin.**

I. Introduction

MULTI-EXCITATION fluorescence hyperspectral imaging (ME-HSI) illuminates a sample at several excitation wavelengths and records, for each, an emission spectrum in many bands. Every pixel thus carries an excitation–emission matrix (EEM) and an image is a four-dimensional array over excitation, emission, height and width. The joint domain separates fluorophores that a single excitation cannot, which is why endogenous autofluorescence from NADH, FAD, collagen and elastin has been used for label-free imaging of tissue state [1]–[4], including cardiac ablation lesions [5]–[7] and tissue classification with ME-HSI [8]–[10]. A particularly important derived quantity is the optical redox ratio, built from the FAD and NADH signals, which tracks cellular metabolism [2], [11].

Tissue is rarely imaged bare. In many applications the fluorophore of interest sits under a cover that scatters, absorbs and fluoresces itself: a collagen-rich layer, an epithelium, a blood-perfused dermis. The cover attenuates the excitation on the way in and the emission on the way out, adds its own autofluorescence, and, because haemoglobin absorbs strongly in the violet and green, *distorts the emission spectrum* rather than merely scaling it [12], [13]. A ratio such as FAD/(FAD+NADH) is therefore not a pure property of the metabolic state. Quantifying how much of its variance is caused by the cover, and designing acquisitions and estimators that are robust to it, requires ground truth that real measurements cannot supply: the true thickness, concentrations and optical properties of every pixel.

Simulation is the natural remedy, and it serves three purposes. (i) It supplies *ground truth* for developing and benchmarking processing methods. (ii) It supports *experiment design*, i.e. choosing excitation lines, emission bands, exposure and geometry before instrument time is spent; each added excitation multiplies acquisition time and photobleaches the sample. (iii) It enables *inverse problems*: parameter retrieval, unmixing and uncertainty quantification with a forward model that is faithful enough to be inverted.

A. Gaps

Photon-transport codes are mature [14]–[17], but they are solvers: they return fluence, reflectance or absorption for a given set of optical properties. A user who wants to simulate an *imaging experiment* must still add the excitation source and its line shape, dye photophysics (inner filter, quenching, FRET, bleaching, Raman), the emission transport, the tunable-filter bands and the detector noise, and then wrap everything in a design-of-experiments and inference pipeline. Hyperspectral scene simulators for remote sensing [18] cover the camera and scene but not fluorescence transport in turbid media. Our previous simulator [19] generated labelled 4D fluorescence cubes with a one-dimensional transport model; it could not represent a cover whose thickness varies laterally, light spreading sideways, or buried inclusions, and it offered no study or inference tools.

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B. Contributions

1) **A schema-first, end-to-end ME-HSI simulator** with a 3D scene editor, a headless Python API and a command-line interface that share one typed experiment contract (Section III).
2) **A physically comprehensive forward model**: ray optics with Fresnel/Snell and microfacet glint, dye photophysics, white-Monte-Carlo transport kernels with reciprocity, and a layered-block 3D Monte Carlo with layer-conforming tallies, a peel-off estimator with multiple-importance weights, voxelised inclusions, and a literature tissue-optics model (Section IV).
3) **A research toolkit**: a study engine with provenance, Sobol'/Morris sensitivity, unmixing, surrogate maximum-likelihood retrieval, Fisher-information/CRLB analysis with D-optimal channel selection, and calibration of a digital twin (Section V).
4) **A validation suite** against analytic and published references, and a sim-to-real study with measured collagen and droplet data (Section VI).
5) **A set of simulation experiments** on collagen-covered tissue that quantify attenuation, spectral distortion, confounding of the redox ratio, inclusion detectability, information-optimal acquisition, depth-dependent unmixing failure and performance (Section VII).

II. Related Work

Tissue-optics Monte Carlo. MCML [14] established the standard for multilayered tissue with implicit capture, Henyey–Greenstein scattering [20] and Fresnel boundaries; the same ideas are reviewed in [21], [22], and voxelised variants such as mcxyz extend them to heterogeneous media. GPU implementations followed, for layered media [23] and for voxel media [15]. Complex boundaries are handled by mesh-based codes [16], [17], [24], [25] and hybrid mesh–voxel methods [26]. Our engine is closest to MCML in its formulation (Section IV-F) but is extended to tilted planar interfaces, voxel inclusions and an excitation/emission two-pass structure with a next-event estimator; it is not a general-purpose replacement for mesh-based codes.

Fluorescence Monte Carlo. Fluorescence propagation was modelled by Welch *et al.* [27] and Jacques [28]; accelerated schemes for layered tissue [29] and time-resolved fluorescence [30] reduce the cost by reusing the excitation transport. We follow the same decomposition, but add quantitative dye photophysics and the full camera chain, and we resolve photobleaching per voxel across an excitation sequence.

Hyperspectral scene simulators. DIRSIG [18] is the best-known physics-based image simulator for remote sensing, with a rich sensor model but without fluorescence in turbid media. Domain randomisation [31] uses synthetic data for sim-to-real transfer in robotics; we instead calibrate a physical model to measured data.

Fluorescence EEM analysis and unmixing. EEM data are classically analysed with multi-way models such as PARAFAC [32], [33]; in imaging, linear unmixing of spectral images is established [34]–[36], with constrained solvers [37], [38] and endmember extraction by NMF [39] or vertex component analysis [40]. Tunable-filter spectral imaging is reviewed in [41]–[43].

Intrinsic fluorescence in turbid media. Wu *et al.* [12] and Müller *et al.* [13] showed how scattering and absorption distort measured fluorescence relative to the intrinsic spectrum, and Richards-Kortum and Sevick-Muraca [44] reviewed quantitative tissue spectroscopy. Our simulator reproduces such distortions from first principles, so that correction strategies can be tested against known ground truth.

Global sensitivity analysis. Variance-based Sobol' indices [45]–[47] and Morris screening [48], [49] are standard in computational modelling. We expose them as first-class operations on simulated imaging experiments and report the Monte Carlo noise floor of the indices.

Experiment design and band selection. Optimal experimental design maximises functionals of the Fisher information matrix [50]; greedy maximisation of monotone submodular criteria has a $1-1/e$ guarantee [51]. In hyperspectral imaging, band selection is usually data-driven [52]–[55]. Here the forward model is known, so channel selection can be posed as a model-based design problem and compared with data-driven selectors.

Digital twins. Digital twins couple a model with measurements of a specific physical asset [56], [57]. We use the term in this restricted sense: a forward model whose instrument and fluorophore parameters are calibrated to laboratory data, with explicit statements of what is and is not identifiable.

III. System

SpectralForge Studio consists of a typed experiment contract, a Python engine and a web application (Fig. 1).

Schema-first contract. An experiment is a single Pydantic object holding fluorophores, materials (each a uniform optical medium), objects, a background, light sources with excitation sequences, cameras and acquisition settings including the physics toggles. The schema is exported to JSON Schema and TypeScript, so the browser editor builds only experiments that validate in the engine. The experiment is embedded verbatim in every exported cube, together with a provenance record (package and library versions, git commit and dirty flag, seed and the SHA-256 hash of the canonical experiment JSON), so any result can be traced to its configuration.

Engine. The engine is written in Python with numba-compiled kernels. A capture first traces camera rays through analytic solids into a wavelength-independent hit buffer, then assembles the spectral radiance for each excitation step from per-material optics and the transport kernel, projects it onto the camera bands and applies the detector model. Layered blocks bypass the 1D kernel and are rendered by the 3D Monte Carlo of Section IV-F. Every random draw has its own seeded stream, and results are bit-identical for a given seed and independent of the thread count.

Web editor and headless access. The browser application provides the 3D scene editor, a live preview, an *Explain pixel* tool that decomposes the radiance of one pixel into named components (dye, secondary, elastic, glint, Raman,

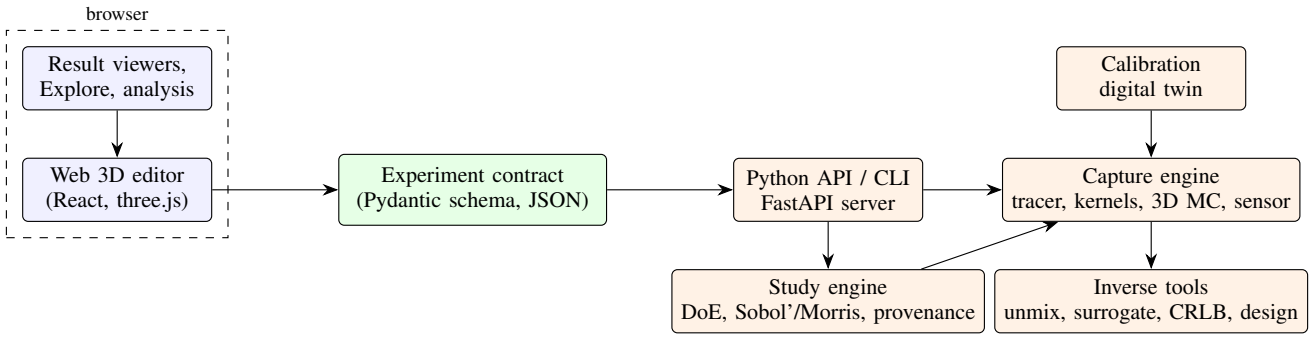


Fig. 1. Architecture of SpectralForge Studio. The experiment contract is defined once in Python and exported as JSON Schema and TypeScript types for the browser, so the editor and the engine cannot diverge. [TODO: replace the sketch with the final architecture diagram]

TABLE I
NOTATION.

Symbol	Meaning
$\lambda_{\mathrm{ex}}, \lambda_{\mathrm{em}}$	excitation / emission wavelength (nm)
$\mu_a, \mu_s, \mu_s', \mu_t$	absorption, scattering, reduced scattering, transport ($\mu_a + \mu_s$) coefficients (mm^{-1})
$g,\ n$	Henyey–Greenstein anisotropy, refractive index
$\varepsilon_k,\ c_k$	molar extinction ($\mathrm{M}^{-1}\mathrm{cm}^{-1}$), concentration of dye k
$\Phi_k,\ e_k(\lambda)$	quantum yield, unit-integral emission density (nm^{-1})
E	excitation irradiance at the surface ($\mathrm{mW\,mm}^{-2}$)
G	transport kernel ($\mathrm{mm\,sr}^{-1}$)
L	radiance ($\mathrm{mW\,mm}^{-2}\mathrm{sr}^{-1}\mathrm{nm}^{-1}$)
$w,\ \Phi_v$	photon-packet weight (mW); track-length fluence in voxel v
$B,\ S$	blood volume fraction, haemoglobin oxygen saturation
c	index of a channel (λ_{ex}, band); C the number of channels
$\boldsymbol{\theta}$	parameter vector (thickness, concentration, ...)
$\mu_c(\boldsymbol{\theta}),\ \sigma_c^2$	expected signal and variance of channel c (e^-, e^{-2})
$\mathbf{F},\ \mathbf{J}$	Fisher information matrix, Jacobian $\partial\mu_c/\partial\theta_i$
$S_i,\ S_{T_i}$	first-order and total Sobol' indices
$\mathbf{E},\ \mathbf{a}$	endmember matrix, abundance vector

leakage, background), and an analysis workspace. The same functionality is available through a headless Python API and a command-line interface, with study endpoints for design of experiments and sensitivity analysis. A built-in mock engine keeps the browser usable when no engine is reachable.

IV. Physics Model

Table I lists the notation. Lengths are in mm, wavelengths in nm, powers in mW and concentrations in µM unless stated.

A. Radiometric formulation

For an excitation irradiance E on a surface, the fluorescence radiance leaving towards the camera is

$$L(\lambda_{\mathrm{em}}) = E \sum_k \langle \mu_a^{(k)} \rangle_{\mathrm{line}}\, \Phi_k\, e_k(\lambda_{\mathrm{em}})\, G(\lambda_{\mathrm{em}}), \qquad (1)$$

where $\langle \mu_a^{(k)} \rangle_{\mathrm{line}}$ is the absorption of dye k averaged over the excitation line shape and

$$\mu_a^{(k)}(\lambda) = \frac{\ln 10}{10}\, \varepsilon_k(\lambda)\, c_k \quad [\mathrm{mm}^{-1}], \qquad (2)$$

with c_k in M and ε_k in $\mathrm{M}^{-1}\mathrm{cm}^{-1}$. The kernel G absorbs everything else: Fresnel transmission in and out, refraction, the $1/n^2$ basic-radiance factor, the $1/4\pi$ of isotropic emission, depth integration, excitation attenuation and emission reabsorption. Inner-filter and reabsorption effects are therefore not tunable parameters; they emerge from G when μ_a is large.

B. Ray optics and material optics

Camera rays are traced through analytic solids (box, sphere, cylinder, droplet, vessels by CSG, and layered blocks); a ray crosses a chain of clear media (glass, liquid) by Snell refraction with unpolarised Fresnel transmission until it meets a turbid medium. Shadow rays from every entry point to the source aperture collect Fresnel factors and Beer–Lambert attenuation of the clear media in front. Specular glint at the first interface follows a microfacet BRDF [58] with Smith shadowing [59] and unpolarised Fresnel reflectance. For side-lit clear media (a laser crossing a cuvette) 16 Gauss–Legendre nodes along the refracted view chord, each with its own light-entry face, replace the slab assumption.

Dyes also experience Stern–Volmer self-quenching,

$$\Phi_{\mathrm{eff},k} = \frac{\Phi_k}{1 + K_{\mathrm{sv}} \sum_j c_j}, \qquad (3)$$

and FRET pairs (D, A, k) transfer with a concentration-dependent efficiency

$$E_{\mathrm{FRET}} = \frac{k\, c_A}{1 + k\, c_A}, \qquad (4)$$

which scales the donor emission by $1 - E_{\mathrm{FRET}}$ and adds a sensitised acceptor term (the efficiency is a lumped parameter; there is no spectral-overlap integral). Water Raman scattering at a fixed wavenumber shift of 3400 cm^{-1} follows the excitation line, $1/\lambda_R = 1/\lambda_{\mathrm{ex}} - 3400 \cdot 10^{-7}$ with λ in nm, with a Gaussian band of 480 cm^{-1} FWHM. Fluorescence that is

re-absorbed on the emission path ℓ is re-emitted once and isotropically by the absorbing dye,

$$\begin{aligned}\sec_k(\lambda) = \Big[\sum_{\lambda'} L(\lambda')\big(e^{\tau}-1\big)\tfrac{\mu_a^{(k)}}{\mu_a^{\mathrm{tot}}}\Big] \\ \times\, \Phi_k\, b_k\, \tfrac{\lambda_{\mathrm{abs}}}{\lambda_{\mathrm{em}}}\, e_k(\lambda) e^{-\tau(\lambda)},\end{aligned} \tag{5}$$

with $\tau(\lambda) = \mu_a^{\mathrm{tot}}(\lambda)\ell$ and b_k the bleaching survival.

C. Photobleaching

Each pixel and dye carries a survival fraction s. After an excitation step the absorbed photon dose per molecule is $\sigma\,\Phi_{\mathrm{ph}}\,t_{\mathrm{tot}}$ with $\sigma = \ln(10)\,10^3\varepsilon/N_A$ (cm^2) and photon flux Φ_{ph} at the illuminated surface (with a build-up factor $\mathrm{clip}(1+2\pi\rho, 1, 6)$ for turbid media, ρ the elastic remission), and

$$x = \Phi_b\,\sigma\,\Phi_{\mathrm{ph}}\,t_{\mathrm{tot}}, \qquad s \leftarrow s\,e^{-x}, \tag{6}$$

where Φ_b is the bleaching quantum yield; fluorescence within the step uses the mean survival $s(1-e^{-x})/x$. Acquisition order therefore matters, as in measurements of fluorescein [60], [61]. In the block engine the dose is resolved per voxel.

D. One-dimensional transport kernels

Clear media. For a non-scattering slab,

$$G = \frac{T_{\mathrm{in}}T_{\mathrm{out}}}{4\pi n^2\cos\theta_i'}\,\frac{1-e^{-\kappa L}}{\kappa}, \quad \kappa = \mu_t^{\mathrm{ex}}\frac{\cos\theta_o'}{\cos\theta_i'} + \mu_t^{\mathrm{em}}, \tag{7}$$

with L the view chord, θ_i', θ_o' the refracted excitation and view angles, and T Fresnel transmittances. The limits are the dilute result $G \to T_{\mathrm{in}}T_{\mathrm{out}}L/(4\pi n^2\cos\theta_i')$ and the front-face plateau $1/\kappa$.

Turbid media: white Monte Carlo and reciprocity. Helmholtz reciprocity equates the radiance escaping at cosine μ_o from an isotropic emitter at depth z with the fluence produced at z by a collimated beam entering at μ_o. Both legs therefore use one family of collimated-incidence fluence profiles f:

$$G(\lambda_{\mathrm{em}}) = \frac{T_{\mathrm{in}}T_{\mathrm{out}}}{4\pi n^2}\int_0^d f_{\mathrm{ex}}(z)\,f_{\mathrm{em}}(z)\,\mathrm{d}z. \tag{8}$$

The white Monte Carlo [62] walks photons with $\mu_s = 1$ and no absorption and histograms the path length s; absorption enters afterwards as the weight $\exp[-(\mu_a/\mu_s)s]$. The baked tables therefore depend only on (g, n, n_{out}) and dimensionless variables, not on wavelength, dye or concentration, and are cached on disk. Per material and excitation, a small table $J(\mu_i', \mu_o', d, \lambda_{\mathrm{em}})$ is assembled by matrix products over path length, and the per-pixel kernel is a fused numba kernel. The same tables yield the elastic remission $\rho(\theta_i, \theta_o)$ (a Lambertian reflector of reflectance R has $\rho = R/\pi$), the mean fluorescence origin depth used by Eq. (5) and a radial subsurface-spread profile. Below $\tau = \mu_s d = 0.02$ a single-scattering thin-slab profile makes the kernel tend smoothly to Eq. (7).

E. Tissue optics

Tissue absorption follows Jacques [63] (eq. 1):

$$\begin{aligned}\mu_a(\lambda) = B\,S\,\mu_a^{\mathrm{HbO_2}} + B(1-S)\,\mu_a^{\mathrm{Hb}} + W\mu_a^{\mathrm{water}} \\ + F\mu_a^{\mathrm{lipid}} + M\mu_a^{\mathrm{mel}} + \sum_k \mu_a^{(k)},\end{aligned} \tag{9}$$

with B the blood volume fraction, S the oxygen saturation, W the water, F the lipid and M the melanosome volume fraction. The haemoglobin terms are $\mu_a^{\mathrm{HbO_2}} = \ln(10)\,\varepsilon_{\mathrm{HbO_2}}C_{\mathrm{Hb}}/64500$ for whole blood ($C_{\mathrm{Hb}} = 150$ g/L) and the melanosome term is $\mu_a^{\mathrm{mel}} = 519(\lambda/500\,\mathrm{nm})^{-3.5}$ cm^{-1}. Optional bilirubin and β-carotene terms and the dyes of Eq. (2) are added. The tabulated spectra are literature data stored as CSV with citation headers: haemoglobin (Prahl compilation, 250–1000 nm), water (Pope and Fry [64] for 380–727 nm, Segelstein and Hale and Querry [65] elsewhere), lipid [66], and PhotochemCAD pigments (organic-solvent spectra, whose peaks in tissue can be shifted by 10–20 nm). Reduced scattering uses the Mie–Rayleigh form of Jacques (eq. 2),

$$\mu_s'(\lambda) = a'\Big[f_{\mathrm{ray}}\Big(\tfrac{\lambda}{500}\Big)^{-4} + (1-f_{\mathrm{ray}})\Big(\tfrac{\lambda}{500}\Big)^{-b_{\mathrm{Mie}}}\Big], \tag{10}$$

with $\mu_s = \mu_s'/(1-g)$. Intralipid phantoms use the measured $\mu_s(\lambda)$ and $g(\lambda)$ of van Staveren *et al.* [67] and fat emulsions are compared with [68]. The library provides 18 materials (three epidermis melanin levels, papillary and reticular dermis, subcutis, muscle, oral and colon mucosa, a dysplastic nodule, liver, breast, brain grey matter, collagen gel and dense collagen, whole blood, an Intralipid phantom and an India-ink absorber) and three tissue presets (a three-layer skin, mucosa with a buried nodule, and a vessel under collagen). Fig. 2 shows the library.

F. Layered-block 3D Monte Carlo

A *layered block* is a stack of layers whose interfaces are exact planes $y = a_i + b_i x + c_i z$ (uniform or linear-wedge thickness) bounded by four vertical faces and the bottom $y = 0$, with air outside. Packets carry power w (mW) and, as in MCML [14], undergo implicit capture $w \leftarrow w\mu_s/\mu_t$, Henyey–Greenstein scattering [20], Russian roulette and unpolarised Fresnel reflection or refraction at the true plane normal, so a wedge steers light correctly and total internal reflection is included. The bottom face is a Fresnel interface to air: $w(1-R)$ leaves and is lost to the stage, wR stays in the block. Absorption is tallied on a *layer-conforming* voxel grid with lateral cells of size Δ and a fixed number of sublayers across the local thickness $t_l(x,z)$, so a voxel volume is $V_v = \Delta^2 t_l/S$ (exact because t_l is linear) and a zero-thickness wedge end is represented without staircasing.

Excitation pass. Once per excitation line, packets are launched from a jittered grid on the faces that face the source with weight $S\cos\theta_{\mathrm{inc}}\,dA\,T_{\mathrm{other}}$ per mW. The specular part of the entry Fresnel split is not transported (the tracer renders it as glint). The pass tallies the track-length fluence

$$\Phi_v = \frac{1}{V_v}\sum_{\text{segments}} w\,\ell, \qquad P_{\mathrm{abs},v} = \mu_a\Phi_v V_v, \tag{11}$$

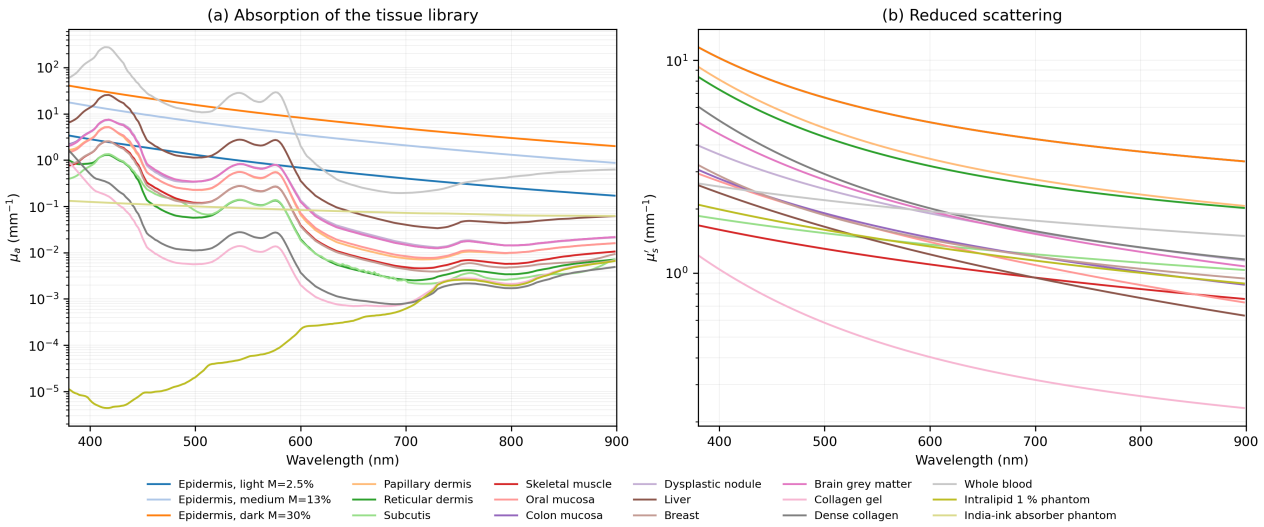


Fig. 2. Literature tissue-optics library: (a) absorption and (b) reduced scattering of the 18 materials, 380–900 nm.

and the energy balance $R + T + A + \text{specular} + \text{lost} = 1$. The absorbed power of a layer is split between its dyes in proportion $\mu_a^{(d)}/\mu_a^{\text{tot}}$; each channel (excitation, layer, dye) becomes an emission source $A_d\,\Phi_d^{\text{eff}}\,s\,(1 - E_{\text{FRET}})$, plus a Raman channel in aqueous layers.

Emission pass. Once per emission wavelength node, source voxels are drawn by stratified inverse-CDF sampling on the total source $q(v) = \sum_c \sigma_{c,v}$, uniformly inside the voxel and with isotropic direction. A packet carries $\text{coef}_c = \sigma_{c,v}\,Q/(q(v)N)$, so a single run serves all channels of all excitations; the radiance tally is then multiplied by the area-normalised emission density $e_c(\lambda_{\text{em}})$. Nodes are placed adaptively (dense where the transport factor bends) and the factor is interpolated linearly between them, since the sharp structure lives in e_c.

Cone estimator and peel-off. Counting only packets that leave inside the camera cone of half-angle θ_a gives the unbiased estimate

$$L_{\text{cell}} = \frac{\sum_{\text{cone}} w}{A_{\text{cell}}\,(\mathbf{n}\cdot\mathbf{c})\,\pi\sin^2\theta_a}, \tag{12}$$

which is noisy for the small cells needed to resolve a wedge. The *peel-off* (next-event) estimator instead adds, at every scattering event and at the launch of an emission packet, the expected contribution of the straight line of sight to the camera. The internal direction $\mathbf{d}$ follows from inverting Snell's law through the interface chain, and

$$\begin{aligned} \delta L_{\text{cell}} &= \frac{w\,p_{\text{HG}}(\mathbf{u}\cdot\mathbf{d})\,T\,e^{-\tau}J}{A_{\text{cell}}(\mathbf{n}\cdot\mathbf{c})}\,\omega_{\text{MIS}}, \\ J &= \prod_j \frac{n_{\text{up},j}^2\cos\vartheta_{\text{up},j}}{n_{\text{in},j}^2\cos\vartheta_{\text{in},j}}, \end{aligned} \tag{13}$$

where τ is the optical depth of the straight segments, T the product of Fresnel transmittances, J the étendue factor that converts solid angle inside to solid angle in air, and $\omega_{\text{MIS}} = 1/(1 + \Omega_c\,p\,J)$ a balance-heuristic multiple-importance weight [69] that bounds the forward-peaked tail of the phase function (Ω_c the cone solid angle). A flag marks the paths the peel-off does not cover (the launch and the path after a Fresnel reflection), which the cone estimator still scores, so nothing is counted twice; side faces use the cone estimator. The peel-off needs about five times fewer packets (photon counts are scaled by 0.2), and the top-face tally is smoothed with a sub-cell Gaussian of $\sigma = 0.35$ mm.

Determinism. Photons are cut into fixed batches that re-seed from (seed, batch) and own private tallies summed in batch order, so results are bit-identical for any thread count.

G. Inclusions

A block may embed spheres, ellipsoids, cylinders and boxes. The layer planes remain exact; inclusions live on a regular Cartesian voxel map over the block's bounding box (voxel size h, eight analytic supersamples per voxel, region id 0 for the layer material and $r \geq 1$ for inclusion r). A later inclusion wins overlaps, a mixed voxel takes the majority id among its in-block samples (ties go to the inclusion) and is optically homogeneous, and inclusions are index-matched to the host (no Fresnel boundary). Host tallies and inclusion tallies are kept separate and together conserve the absorbed power. Inclusions can themselves carry dyes and emit. A block without inclusions takes exactly the earlier code path (bit-identical results).

H. Sensor, tunable-filter bands and noise

The camera model follows a thin lens focused at distance S with f-number N, transmission T, magnification m. For a pixel of area A (including binning) exposed for t, the electrons collected in band b are

$$\begin{aligned} e_b &= k_0\cos^4\theta\,A\,t\int L(\lambda)R_b(\lambda)\,T_{p,b}\,\text{QE}(\lambda)\frac{\lambda}{hc}\,\text{d}\lambda, \\ k_0 &= \frac{\pi T}{4N^2(1+m)^2}, \end{aligned} \tag{14}$$

with R_b the band response (pass-band times long-pass edge) and $T_{p,b}$ the peak transmission. Liquid-crystal tunable filters

(LCTF) [41] are Gaussian pass-bands, filter wheels super-Gaussian of order six; a ray at field angle θ sees the centre wavelength shifted to $\lambda_0\sqrt{1-\sin^2\theta/n_{\text{eff}}^2}$, and the cube is resolved in six field-angle rings. Blur is the thin-lens circle of confusion with an Airy floor and axial chromatic shift, varying per pixel and band. The detector chain is

$$\begin{aligned} S \to S(1+\text{PRNU}) \to +\text{dark} \to \text{Poisson} \to +\mathcal{N}(0,\sigma_r^2) \\ \to \text{clip} \to \div g \to \text{round}, \end{aligned} \tag{15}$$

so for one pixel the variance in electrons squared is

$$\sigma^2(S) = (S+D) + \sigma_r^2 + g^2/12, \tag{16}$$

where D is the dark signal and $g^2/12$ the quantisation noise [70], [71]; for the mean of an ROI with n pixels it is divided by n. A white-reference mode renders a Spectralon plate with the same band set so that sample/white reproduces the laboratory normalisation, and a dark mode removes the source.

V. Research Toolkit

A. Study engine and provenance

A `StudySpec` names a base experiment (or preset), a list of parameters addressed by JSON pointers (continuous, integer or choice, linear or log scale, with optional linked pointers), a design (grid, random, Latin hypercube, Sobol' sequence, Saltelli or Morris), replicate seeds and metrics (ROI band sums and ratios, peak wavelength, EEM integral, Michelson or Weber contrast, contrast-to-noise ratio, ground-truth statistics). Each run is journaled so a study is resumable, every capture can embed its provenance, and a Methods-text generator produces a description of the experiment and study from the specification.

B. Global sensitivity analysis

For inputs scaled to the unit cube and a Saltelli design with matrices A, B and AB_i (A with column i from B) and N base samples, the first-order and total effects are

$$V_i = \frac{1}{N}\sum_j f(B)_j\big[f(AB_i)_j - f(A)_j\big], \tag{17}$$

$$V_{T_i} = \frac{1}{2N}\sum_j \big[f(A)_j - f(AB_i)_j\big]^2, \tag{18}$$

with $S_i = V_i/\mathrm{Var}(Y)$ and $S_{T_i} = V_{T_i}/\mathrm{Var}(Y)$ (estimators of Saltelli *et al.* [46] and Jansen [47]; Sobol' indices [45]). Confidence intervals are percentile bootstraps over base rows. Morris screening [48], [49] reports $\mu^* = \mathbb{E}|EE_i|$ and $\sigma = \mathrm{sd}(EE_i)$ of the elementary effects. With R replicate seeds per design point, the within-point variance s_w^2 gives the Monte Carlo noise floor $(s_w^2/R)/\mathrm{Var}(\text{replicate means})$, the fraction of output variance that is engine noise; indices below it are not distinguishable from zero.

C. Unmixing

The joint EEM of a pixel (all bands of all excitations, C channels) is modelled as $\mathbf{y} = \mathbf{E}\mathbf{a} + \boldsymbol{\epsilon}$, which is exact for independent emitters seen through the same optical path and approximate when the paths differ, which is precisely the cover effect studied here. Endmembers come from the forward model (the per-pixel component decomposition at the purest pixels, an in-silico equivalent of a pure-component capture through the same instrument), from user ROIs, or from NMF [39] or VCA [40]. Solvers are least squares, non-negative least squares [38] and fully constrained least squares ($a_k \ge 0,\ \sum_k a_k = 1$) by the augmented-NNLS construction of Heinz and Chang [37], optionally with inverse Poisson–Gaussian channel weights.

D. Surrogates and maximum-likelihood retrieval

A surrogate replaces the Monte Carlo forward model by an interpolant $\mu_c(\boldsymbol{\theta})$ of the noise-free ROI-mean signal of every channel on a grid or Sobol' set of one to three parameters. The noise-free output (expected electrons, before detector noise) is used, and the Monte Carlo seed is fixed across the grid (common random numbers) so finite-difference Jacobians are smooth; residual Monte Carlo noise is removed with a smoothing spline chosen by generalised cross-validation. The estimator minimises

$$\begin{aligned} \chi^2(\boldsymbol{\theta}) &= \sum_c \frac{[y_c - \mu_c(\boldsymbol{\theta})]^2}{\sigma_c^2(\mu_c)}, \\ \sigma_c^2 &= \frac{(\mu_c + D_c) + \sigma_r^2 + g^2/12}{n}, \end{aligned} \tag{19}$$

by iteratively re-weighted trust-region least squares, with start points from a scan of a dense surrogate table so that non-monotonic signals do not depend on the user's guess. The fixed point solves the Gaussian maximum-likelihood and, for Poisson data, the Poisson likelihood estimating equation $\mathbf{J}^\top\mathbf{W}(\mathbf{y}-\boldsymbol{\mu}) = 0$. Confidence intervals are profile-likelihood intervals [72], $\{\theta_p : \min_{\text{others}}\chi^2 - \chi^2_{\min} \le q_{\chi^2}(\text{level},1)\}$, and open intervals are flagged. A map retrieval fits one parameter vector per tile of a noisy capture and reports bias, RMSE and interval coverage against the ground truth.

E. Fisher information, CRLB and optimal design

With channels independent and Poisson–Gaussian variance (16), the Fisher information is

$$\mathbf{F} = \sum_{c=1}^{C}\mathbf{I}_c, \qquad \mathbf{I}_c = \frac{\mathbf{J}_c\mathbf{J}_c^\top}{\sigma_c^2(\mu_c)}, \quad J_{c,i} = \frac{\partial\mu_c}{\partial\theta_i}, \tag{20}$$

and the Cramér–Rao bound on any unbiased estimator is $\mathrm{Cov}(\hat{\boldsymbol{\theta}}) \succeq \mathbf{F}^{-1}$. Eq. (20) equals the Poisson information μ'^2/μ when read noise vanishes and the Gaussian information when the signal is read-noise limited; the term from the dependence of the variance on $\boldsymbol{\theta}$ is neglected. Because $\mathbf{F}$ is a sum of per-channel matrices, choosing k channels is the classical design problem [50]. D-optimality maximises $\log\det\mathbf{F}$, A-optimality minimises $\mathrm{tr}\,\mathbf{F}^{-1}$, and the target (c-optimal) criterion minimises $[\mathbf{F}^{-1}]_{pp}$, the bound of one parameter. The first is

monotone and submodular, so the greedy set is within $1-1/e$ of the optimum [51]; it is checked against exhaustive search when $\binom{C}{k}$ is small. Exposure allocation over selected channels uses $I_c(t) = \mathbf{J}_c\mathbf{J}_c^\top (t/t_0)^2/\sigma_c^2(t)$ under a total-time budget and a full-well limit: it is convex in the shot-limited regime and solved by sequential quadratic programming otherwise. The selected set can be compared with the channels chosen by the label-free selectors of our earlier work [55].

F. Calibration and digital twin

For an optically thin, dilute layer the electrons in band b at excitation λ_{ex} are

$$S[\lambda_{\mathrm{ex}}, b] = \sum_k a_k\, A_k(\lambda_{\mathrm{ex}})\, M_k(\lambda_{\mathrm{ex}}, b), \tag{21}$$

with $A_k = \sum_\lambda \mathrm{line}_{\mathrm{ex}}(\lambda)\,\mathrm{exc}_k(\lambda)$ the excitation overlap, $M_k = \sum_\lambda R_b(\lambda) e_k(\lambda)$ the emission band projection and $a_k = c_k \varepsilon_k \Phi_k \ln 10/10$ a lumped amplitude that cannot be separated into concentration, extinction and yield from relative data. Gains (a global scale, or one scale per excitation) are eliminated in closed form by variable projection [73], so the optimiser sees only shape parameters (excitation and emission peak and width, red-tail skew, vibronic shoulder). The objective is a robust soft-L1 loss of per-excitation relative residuals from deterministic multi-starts; covariance is $s^2(\mathbf{J}^\top\mathbf{J})^+$ (statistical only), and a pixel bootstrap is available. Optional shared instrument terms are a smooth log-throughput polynomial and an LCTF ripple orthogonal to cubic polynomials. Which parameters are identifiable is reported explicitly, e.g. an emission peak below the first band is not.

VI. Validation

All numbers below are regenerated by scripts in the repository; only literature reference values are hard-coded. Table II summarises the checks; the following subsections give details.

A. Monte Carlo transport against standard benchmarks

The baked white-Monte-Carlo tables and a conventional MCML implementation both reproduce the Wang–Jacques/Prahl slab benchmark ($\mu_a = 1$, $\mu_s = 9$ mm^{-1}, $g = 0.75$, $n = 1$, $d = 0.2$ mm; $R_d = 0.09734$, $T_t = 0.66096$) to better than $0.1\,\%$ (Fig. 3), conserve energy to 10^{-9}, match the exact incoherent multiple-reflection solution for a non-scattering, index-mismatched slab to $0.01\,\%$ and agree with each other for an oblique, index-mismatched turbid slab to $\leq 0.05\,\%$. Semi-infinite diffuse reflectance agrees with the Farrell–Patterson albedo formula [74] to $0.3\,\%$ ($n = 1$) and $1.1\,\%$ ($n = 1.4$; the dipole formula itself is good to a few percent at an index step). For a high-albedo medium ($\mu_a = 0.01$, $\mu_s' = 1$ mm^{-1}, $g = 0.9$) the absorbed-weight profile decays as $\exp(-\mu_{\mathrm{eff}} z)$ with a fitted $\mu_{\mathrm{eff}} = 0.1736$ mm^{-1} against $\sqrt{3\mu_a(\mu_a + \mu_s')} = 0.1741$ mm^{-1} ($-0.29\,\%$) over nine decades [75].

case	quantity	engine	reference	difference	reference source
MCML slab (mua=1, mus=9, g=0.75, n=1, d=0.2 mm)	Rd	0.09742	0.09734	+0.08 %	Wang & Jacques 1995 / Prahl
	Tt	0.66090	0.66096	-0.01 %	
	R+T+A	1.00000	1.00000	+0.00 %	energy conservation
Baked 'white' tables, same slab	Rd	0.09736	0.09734	+0.02 %	as above
	Tt	0.66137	0.66096	+0.06 %	
Index-mismatched clear slab (n=1.5, mua=0.8/mm, d=1 mm)	Tt	0.41426	0.41424	+0.01 %	incoherent multiple reflections, exact
	Rd	0.00744	0.00745	-0.06 %	
White tables vs conventional MCML (n=1.4, oblique, turbid)	Rd	0.16519	0.16525	-0.04 %	independent code path
	Tt	0.23807	0.23804	+0.01 %	
	A	0.59655	0.59670	-0.03 %	
Semi-infinite diffuse reflectance, a'=0.99, n=1.0	Rd	0.74465	0.74671	-0.28 %	Farrell/Patterson albedo formula
Semi-infinite diffuse reflectance, a'=0.99, n=1.4	Rd	0.61927	0.61277	+1.06 %	Farrell/Patterson albedo formula

Fig. 3. Baked Monte Carlo tables and a conventional MCML code against the Wang–Jacques slab benchmark [14].

B. Optics, photophysics and sensor

Two independent implementations of the clear-medium physics, the closed form of Eq. (7) and the per-pixel volume quadrature, agree to $0.008\,\%$ from 0.1 µM to 1 mM fluorescein for a laterally uniform beam. In the dilute limit the 3D engine and the legacy 2D engine give the same emission shapes and cross-excitation ratios up to one geometry constant (61 points; coefficient of variation $0.09\,\%$). The inner-filter effect behaves as expected [76] (Fig. 4): in front-face geometry the signal per unit concentration saturates, while in 90° geometry the cuvette-integrated signal peaks near 13 µM ($A_{1\,\mathrm{cm}} \approx 0.9$) and the centre-pixel signal falls as $10^{-A/2}$. Reabsorption shifts the apparent fluorescein maximum from 512 nm (dilute) to 540 nm at 500 µM across 9 mm of liquid, but only to 518 nm in the front-face slab. Photobleaching follows $\exp(-kn)$ with $k = 0.094$ per step in a worked case, and the same total dose delivered as 300→40 mW or 40→300 mW records 76/55 % versus 96/71 % of the un-bleached signals. The photon-transfer curve of the detector chain returns the configured gain (1.501 against 1.5 e$^-$/ADU) and read noise (7.99 against 8.0 e$^-$) within $0.2\,\%$ and $0.1\,\%$. The depth-of-field blur coincides with the analytic thin-lens circle of confusion, and a 600 nm band shifts by -1.5 nm at the corner of the lab-replica field ($\theta_{\max} = 8°$, -13 nm at 25°).

C. Layered-block 3D Monte Carlo

Homogeneous slabs reproduce the MCML implementation to Monte Carlo noise (benchmark slab R $-0.21\,\%$, T $+0.04\,\%$; index-mismatched slab R $-0.05\,\%$, T $+0.07\,\%$; tissue-like slab R $-0.55\,\%$, T $+0.98\,\%$, A $-0.54\,\%$), energy is conserved to 10^{-9}, and two identical stacked layers equal one layer of the summed thickness (R $-0.04\,\%$, T $-0.11\,\%$). A 2 mm fluorescent layer reproduces the 1D kernel result of Eq. (1) to $+0.14\,\%$ for fluorescence and $-4.04\,\%$ for elastic remission (Fig. 5). For clear slabs the bottom-boundary treatment matches the closed forms $T = (1-R_0)^2 e^{-\tau}/(1-$

TABLE II
SUMMARY OF VALIDATION CHECKS (ENGINE VERSUS ANALYTIC OR INDEPENDENT REFERENCE).

Check	Reference	Result	Section
Wang–Jacques slab, R_d, T_t	0.09734, 0.66096	$< 0.1\,\%$ deviation	VI-A
Energy conservation (1D and 3D MC)	$R + T + A + \ldots = 1$	10^{-9}	VI-A, VI-C
Clear index-mismatched slab	exact incoherent solution	$0.01\,\%$	VI-A
Diffusion decay, μ_{eff}	0.1741 mm^{-1}	0.1736 mm^{-1} ($-0.29\,\%$)	VI-A
Closed form vs. volume quadrature	internal	$0.008\,\%$	VI-B
Dilute limit vs. legacy engine	ratio CV	$0.09\,\%$	VI-B
Photon transfer: gain, read noise	1.5 e$^-$/ADU, 8.0 e$^-$	1.501, 7.99	VI-B
3D MC vs. MCML (benchmark slab)	MCML	R $-0.21\,\%$, T $+0.04\,\%$	VI-C
3D bottom boundary, clear slab	closed form	agreement to 5 digits	VI-C
Peel-off vs. cone	cone estimator	0.993–1.010	VI-C
Twin fast model vs. full capture	full engine	$4.1 \cdot 10^{-4}$ max. rel. dev.	VI-D

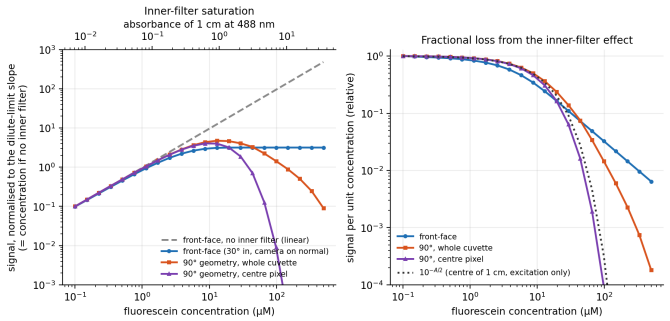


Fig. 4. Inner-filter effect for fluorescein excited at 488 nm: front-face versus 90° geometry. The dotted curve is the excitation-only prediction $10^{-A/2}$ for the cell centre.

$R_0^2 e^{-2\tau}$) and $R_{\mathrm{top}} = (1-R_0)^2 R_0 e^{-2\tau}/(1-R_0^2 e^{-2\tau})$ to the digits reported (e.g. $n = 1.5$, $\mu_a d = 0.5$: $T_{\mathrm{bottom}} = 0.55931$ for both). Turbid fluorescent slabs captured as a block and as a 1D box (secondary emission off) agree: block/1D $= 0.982$ at 6 mm and 0.987 at 1 mm. Thick slabs (about 20 mm) differ by up to $20\,\%$ in the red tail for a physical reason: the 1D engine assumes laterally uniform irradiance while the block sees the finite lamp spot.

The peel-off estimator agrees with the cone estimator within 0.993–1.010 across four configurations (Fig. 6); at equal photon count the cell-to-cell standard deviation of a uniform field is reduced by a factor 3.7 at 10^5 photons and 2.0 at $3 \cdot 10^6$ photons (about $6\times$ for $g = 0$ and $2\times$ for $g = 0.9$), and a sub-cell smoothing of 0.35 mm reaches the noise level of the old cone estimator with five times fewer photons. For a buried fluorescein layer under a wedge cover the signal per unit surface irradiance falls roughly exponentially with local thickness (effective attenuation 0.43 mm^{-1} at 490 nm excitation) and the blue/red emission ratio stays within 1.78–1.83, i.e. this nearly neutral cover attenuates but does not distort (Fig. 7). Under collagen (tissue with NADH/FAD at 365 nm), the buried signal drops by more than an order of magnitude between 0.2 and about 1 mm and the 440 nm band falls less than the 520 nm band, and the cover's own autofluorescence saturates with a decay length of 0.37 mm (Fig. 8).

D. Sim-to-real: calibrating the twin

Machinery. The fast forward model of Eq. (21) agrees with a full capture to a maximum relative deviation of $4.1 \cdot 10^{-4}$ (target $3\,\%$). Fitting noisy full captures (eight seeds, global gain, true emission 470/80 nm, excitation 350/70 nm) recovers the emission peak with bias $+0.02$ nm (sd 0.06 nm) at high signal and $+0.31$ nm (sd 0.63 nm) at low signal, with $95\,\%$ interval coverage of $100\,\%$ for four of five parameters; without dark subtraction the low-signal peak is biased by -1.5 nm.

Collagen. For six measured collagen ROIs (Pepsin and Acetic Acid samples) a one-component twin with per-excitation gain and a smooth throughput correction reaches $R^2 = 0.80$–0.92 and a mean spectral angle (SAM) of 6.2–7.8°, against $R^2 = 0.21$–0.78 and 11.0–16.9° for the nominal instrument (Fig. 9; Pepsin ROI 2: $R^2 = 0.919$, SAM 7.0° against 0.638 and 14.9°). Two components are no better and have a higher BIC. The calibration exposes what is not identifiable: the Nuance window starts at 420 nm, so the emission peak sits on its 380 nm bound; the identifiable descriptors are the centroid above 420 nm (498–505 nm) and the red half-maximum (516–526 nm). The fitted per-excitation gains reveal a radiometric inconsistency of the processed collagen data (a step of about 10^3 between 340 and 365 nm), which a global-gain model would hide. Refitting under alternative instruments (LCTF FWHM 10/14 nm, long-pass 405/420 nm, UV-enhanced CCD) moves the centroid only within 497–499 nm, but an unphysical flat LCTF throughput moves descriptors by 20–50 nm, so the calibrated emission is conditional on the LCTF curve.

Hold-out validation (Fig. 10; mean SAM). B1, calibrate on three excitations and predict the other three (six ROIs, four splits): corrected twin 8.4°, twin with ripple 8.6°, two components 12.3°, nominal 14.2°, empirical pooled spectrum 7.7°, flat 23.7° (in-sample refit 6.4°). B2, calibrate on one ROI and predict another (30 pairs): 7.8° (same dataset) and 7.5° (other dataset), empirical 6.9°/6.7°, flat 23.7°. B3, droplets with global gain, train on 310/340/385/415 nm and predict 325/365/400 nm absolutely without refitting a gain (16 drops): median $R^2 = 0.956$ (nominal 0.951), median SAM 6.0°, median $|\ln(\text{predicted}/\text{measured peak})| = 0.17$. The twin therefore predicts held-out excitations, samples and datasets with SAM about 1–2° above its in-sample value and

case	quantity	3D	reference	diff.	reference
slab benchmark (n=1)	R	0.09718	0.09738	-0.21 %	MCML (transport_mc.mcml_slab)
slab benchmark (n=1)	T	0.6612	0.661	+0.04 %	MCML (transport_mc.mcml_slab)
slab benchmark (n=1)	A	0.2416	0.2417	-0.03 %	MCML (transport_mc.mcml_slab)
slab benchmark (n=1)	R+T+A+spec	1	1	-0.00 %	energy balance
slab index mismatch n=1.4	R	0.08836	0.0884	-0.05 %	MCML (transport_mc.mcml_slab)
slab index mismatch n=1.4	T	0.5274	0.527	+0.07 %	MCML (transport_mc.mcml_slab)
slab index mismatch n=1.4	A	0.3565	0.3568	-0.10 %	MCML (transport_mc.mcml_slab)
slab index mismatch n=1.4	R+T+A+spec	1	1	-0.00 %	energy balance
slab tissue-like	R	0.332	0.3338	-0.55 %	MCML (transport_mc.mcml_slab)
slab tissue-like	T	0.352	0.3486	+0.98 %	MCML (transport_mc.mcml_slab)
slab tissue-like	A	0.2905	0.2921	-0.54 %	MCML (transport_mc.mcml_slab)
slab tissue-like	R+T+A+spec	1	1	+0.00 %	energy balance
fluorescent layer, 2 mm, 60 mm footprint	fluorescence radiance	7.884e-06	7.872e-06	+0.14 %	1-D kernel E mu_a,d Phi e G
	elastic radiance / E	0.07053	0.0735	-4.04 %	1-D kernel rho
two identical layers (0.4+0.6 mm) vs one (1.0 mm)	R	0.2194	0.2195	-0.04 %	single layer
	T	0.1812	0.1814	-0.11 %	
	A	0.5716	0.5713	+0.05 %	

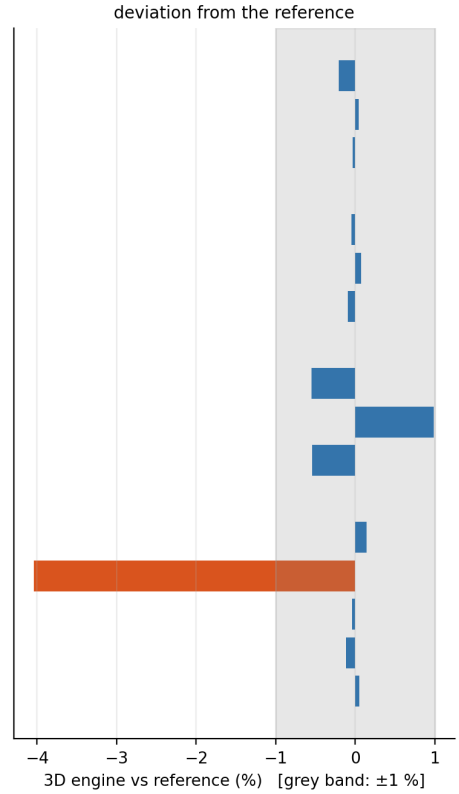


Fig. 5. The 3D Monte Carlo against the MCML implementation and the 1D engine for homogeneous slabs and a fluorescent layer.

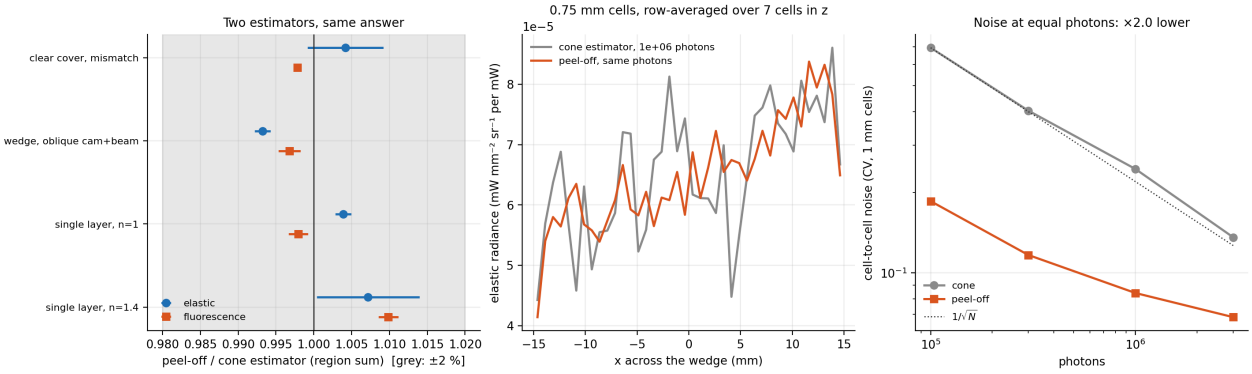


Fig. 6. Peel-off versus acceptance-cone estimator: agreement of region sums and noise reduction at equal photon count.

removes about 90 % of the flat-baseline error, but it does *not* beat a non-parametric copy of another measured spectrum, and the excitation-dependent shape changes of collagen are not captured by one or two components. Its advantage is that it transfers to instrument settings that an empirical copy cannot represent. Scattering of the measured samples is not identifiable: all bands start at least 25–30 nm above the excitation line and there is no flat-field white reference.

VII. Experiments

All experiments use the public API on the tissue-under-collagen scene rebuilt with the literature tissue materials of Section IV-E, run on an Apple M4 Pro (14 cores), with fixed seeds, replicate seeds for Monte Carlo error and bootstrap intervals. Numbers will be taken from the `FINDINGS.md` file of each experiment.

A. E2: Cover-thickness response

Collagen thickness is swept from 0 to 2 mm (at least nine values, three seeds) for each excitation. We (a) fit $I = I_0 \exp(-t/\delta_{\text{eff}})$ to the buried-layer signal and compare δ_{eff} with the diffusion penetration depth $\delta = 1/\sqrt{3\mu_a(\mu_a + \mu_s')}$ of collagen at λ_{ex} and λ_{em}; (b) track the cover autofluorescence fraction from *Explain pixel* and the crossover thickness at which the cover dominates; (c) measure spectral distortion, the emission-peak shift and blue/red ratio against thickness, with blood in the cover ($B = 0, 1, 3$ %) to expose the Soret and Q-band dips; (d) compare the wedge line-profile response with

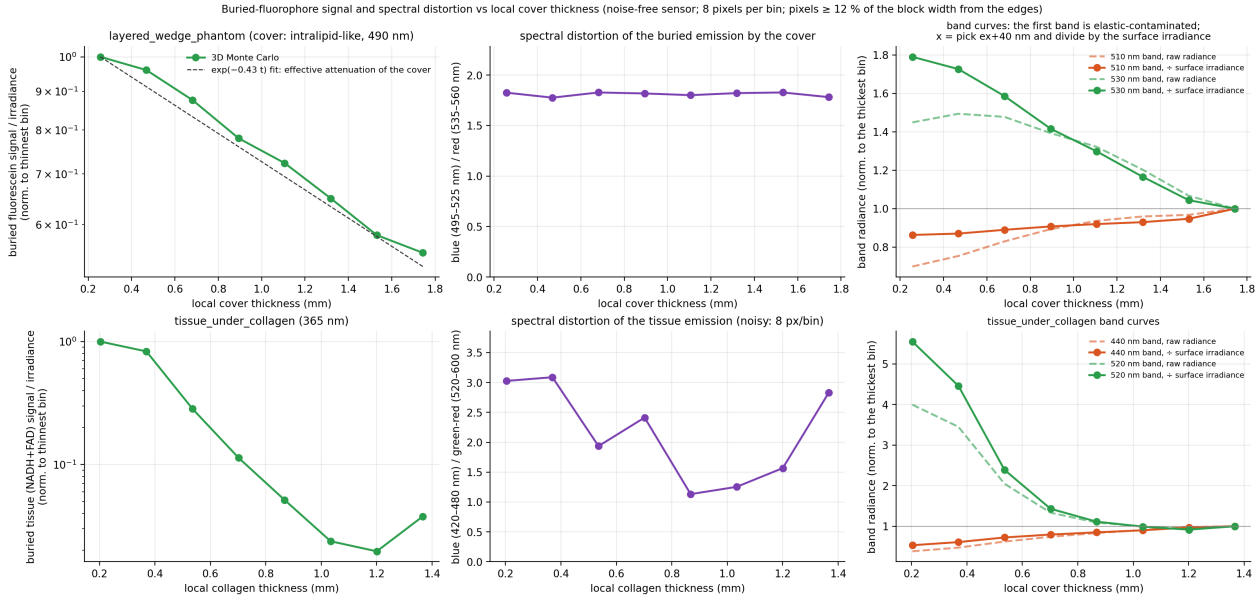


Fig. 7. Buried fluorophore under a wedge cover: signal against local cover thickness, and band ratios for a nearly neutral cover and for tissue under collagen.

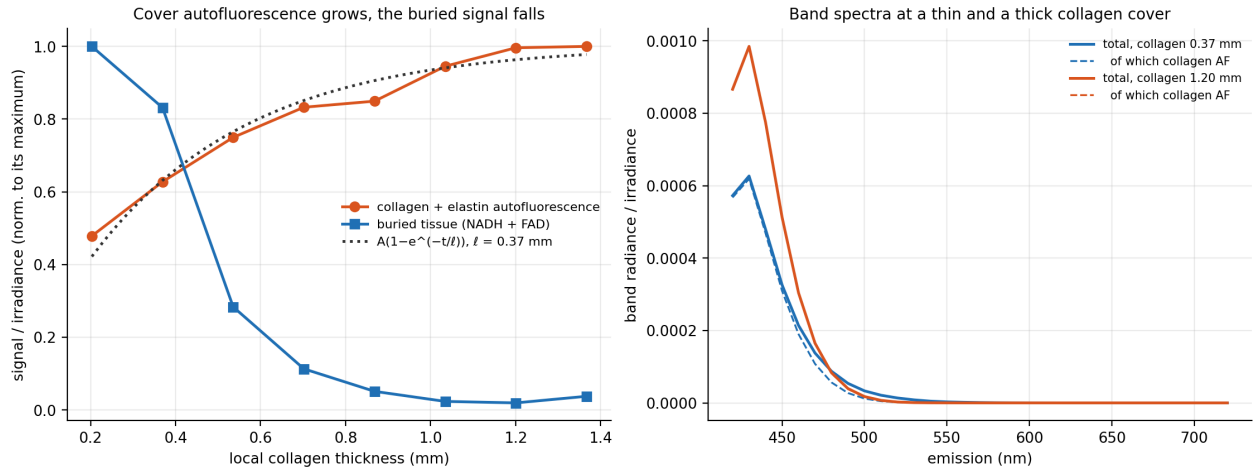


Fig. 8. Collagen cover autofluorescence against thickness (saturating with decay length 0.37 mm) while the buried NADH/FAD signal falls.

the analytic local thickness to separate 3D lateral transport from 1D effects. [TODO: results: δ_{eff} per excitation with CI, crossover thickness, peak shift, wedge blur].

B. E3: Global sensitivity and confounders

Using a Saltelli design ($N \geq 256$ base samples or what completes in 25 min; report N and the noise floor) over six to eight inputs (collagen thickness, $\mu_s'(500)$ and b_{Mie} of the cover, blood fraction, S, buried NADH and FAD concentrations, collagen fluorophore concentration) we report Sobol' S_i, S_{T_i} with bootstrap intervals and Morris μ^*, σ for the NADH band (365/460 nm), the FAD band (450/530 nm), the optical redox ratio FAD/(FAD+NADH) and the peak wavelength. The key question is how much variance of the measured redox ratio is caused by the cover rather than by the true NADH/FAD state; we then test a correction that ratios against an elastic or cover-autofluorescence channel. [TODO: results: share of redox-ratio variance due to cover; indices before/after correction].

C. E4: Inclusion detectability

A spherical fluorescent nodule ($r = 0.5, 1, 2$ mm) at 0.25–3 mm depth and a blood-vessel cylinder ($r = 0.25$–1 mm, depth 0.25–2 mm) lie under collagen. The contrast-to-noise ratio (nodule against a background ROI, with sensor noise) is computed against depth for each excitation, and the maximum detectable depth (CNR = 3) against excitation wavelength and size, interpreted with the effective attenuation. [TODO: results: CNR curves, maximum depth table, image panels].

D. E5: Information-optimal acquisition

On the lab-replica instrument (eight excitations by 31 bands), we compute per-channel Fisher information and the

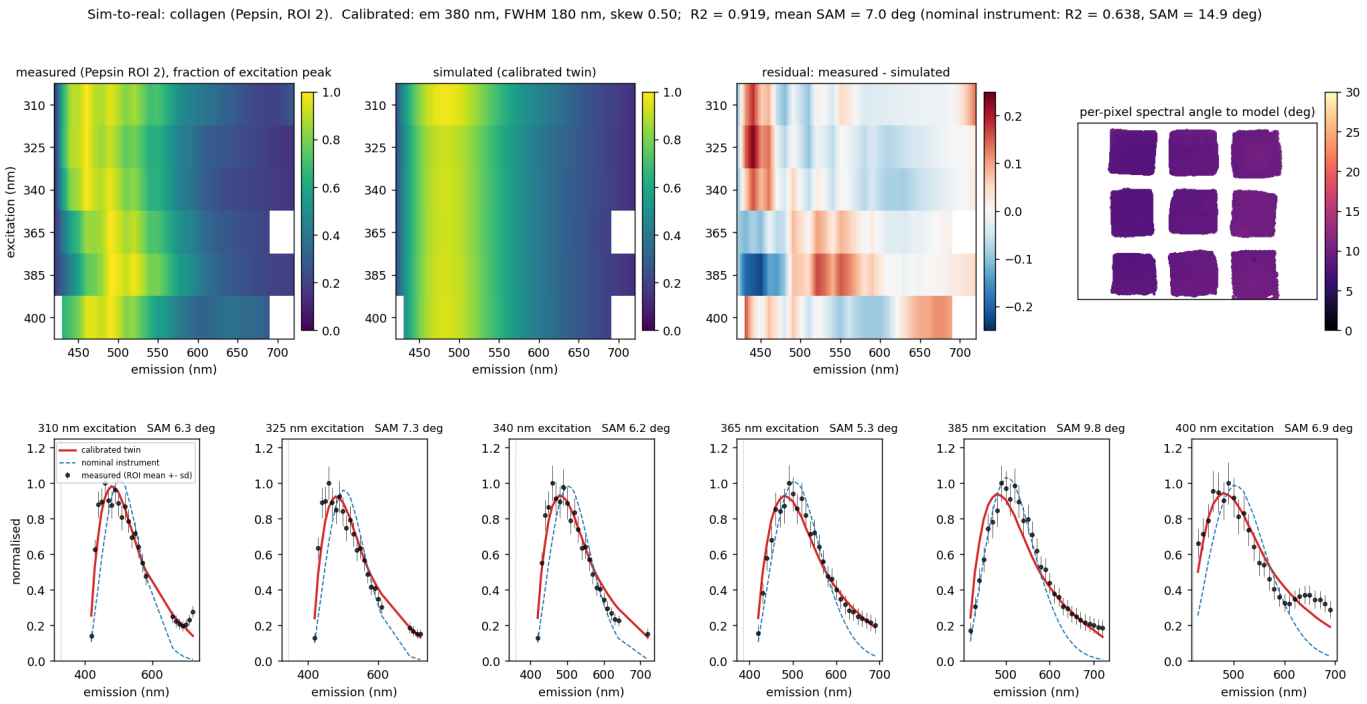


Fig. 9. Calibrated twin for collagen (Pepsin ROI 2): measured and simulated EEM, residual, per-pixel spectral angle and spectra at each excitation (calibrated and nominal instrument).

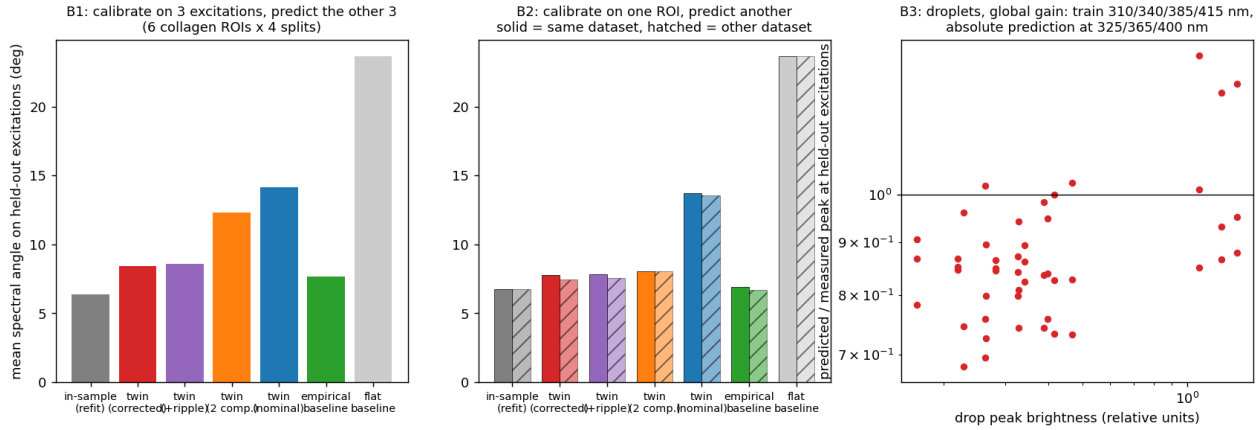


Fig. 10. Hold-out validation. B1: held-out excitations. B2: held-out ROIs. B3: absolute amplitude prediction for droplets at held-out lines.

CRLB for (a) cover thickness and (b) the joint (thickness, buried NADH concentration, cover μ_s'), run greedy D-optimal and target-CRLB selection for $k = 1 \dots 30$, and ask how many channels come within $10\,\%$ of the CRLB of the full protocol and what acquisition time that saves. Baselines are at least 50 random subsets (median, IQR) and the label-free channel selection of our earlier work [55]. The CRLB is checked against the empirical standard deviation of the ML fit (efficiency). [TODO: results: k within $10\,\%$ of full CRLB, time saving, efficiency].

E. E6: Thickness mapping

Per-tile cover thickness is retrieved on the wedge phantom and on a tissue-optics wedge with the surrogate fit; we report bias, RMSE and 95 % interval coverage against three exposure levels, using all channels and the k optimal ones. [TODO: results: bias, RMSE, coverage].

F. E7: Depth-dependent unmixing failure

NNLS unmixing of buried NADH, FAD and collagen abundances with pure-component endmembers is evaluated against cover thickness to quantify the abundance bias caused by spectral distortion, and we test whether joint multi-excitation unmixing or endmembers simulated at the correct depth removes it. [TODO: results: bias against thickness, effect of depth-matched endmembers].

G. E8: Sim-to-real consolidation

The results of Section VI-D are consolidated, and the calibrated collagen fluorophore is placed in a simulated collagen-cover scene whose EEM shape is compared with the measured one (spectral angle per excitation). [TODO: results: spectral angles of the simulated cover scene].

H. E9: Synthetic benchmark for unsupervised clustering

Using the dataset generator we create at least 12 captures of a layered tissue scene with three to four materials or inclusions and cover thickness randomised over 0–1.5 mm. K-means and a Gaussian mixture on PCA features, and one method of the author's repository, are scored by ARI and NMI against ground truth as a function of local cover thickness, with and without the information-optimal channels of E5. [TODO: results: ARI/NMI against thickness; effect of channel selection].

I. E10: Performance and convergence

We report runtime against photon count, the Monte Carlo relative standard error against photon number (expected slope $-1/2$ on log–log axes, fitted), thread scaling (1, 2, 4, 8, 14 threads), peel-off versus cone noise, and agreement of preview with full mode. [TODO: results: slope, scaling table, hardware]. Earlier measurements on a loaded laptop gave 2.3 s for a wedge-phantom preview and 45 s for a six-excitation full capture, and 3.8 s and 61 s for tissue under collagen with eight excitations.

VIII. Discussion and Limitations

What the tool is for. The simulator is most useful where ground truth and controlled variation are needed: quantifying how a cover confounds a spectral ratio, designing channel sets, and benchmarking estimators. The toolkit's model-based channel selection complements data-driven band selection; whether it transfers to real data depends on the fidelity of the twin, which Section VI-D documents honestly, including that a non-parametric spectrum is a competitive predictor.

Physics limitations. Absorption and emission spectra of the dyes are parametric (Gaussians with skew and an optional vibronic shoulder), so there is no $S_0 \rightarrow S_2$ band (fluorescein absorbs nothing at 405 nm in the model against about 3 % of its peak in reality), no exact mirror image and no solvent shift. Scattering is a power law with one Henyey–Greenstein g per material, with no wavelength-dependent g, polarisation or coherence. Fluorescence has a single Stern–Volmer constant, FRET without overlap integral or orientation factor, one-generation isotropic secondary emission, only the 3400 cm^{-1} water Raman band, and no lifetimes, depolarisation, temperature dependence or triplet saturation. Bleaching in the 1D engine uses the surface fluence. Block pixels have no secondary emission (fluorescence re-absorbed in the block is lost, making it a few percent to about 10 % dimmer for strongly overlapping dyes), a single camera direction per block, planar interfaces and a rectangular footprint, and inclusions are index-matched. Tissue pigments measured in organic solvents are proxies for the bound pigments. The bottom of the block is a Fresnel interface to air and the stage never reflects. Indirect diffuse interreflection between objects is not implemented.

Statistical limitations. The 3D result is a Monte Carlo estimate (about 10 % rms per 0.5 mm face cell before smoothing; 2–3 % in a 3 mm region of interest), which is why studies report replicate noise floors. Fisher information is a local bound that assumes an unbiased estimator and independent channels and neglects fixed-pattern noise. Calibration intervals are statistical and the systematic error from the instrument model dominates (Section VI-D). The twin is calibrated on thin, dilute-equivalent samples, so collagen and droplet parameters are effective rather than intrinsic.

Validation scope. The validation is against analytic results, MCML-type benchmarks and one laboratory's measurements; we have not yet compared 3D inclusion or tissue-phantom predictions with independent phantom measurements. [TODO: state any phantom measurement added before submission].

IX. Conclusion

SpectralForge Studio couples a literature-grade layered-tissue 3D Monte Carlo, dye photophysics and a camera model with a study engine, inverse tools, information-theoretic design and calibration into a single, schema-driven, reproducible environment. The transport engine matches established benchmarks to well below one percent, and the calibrated twin predicts held-out excitation lines with a mean spectral angle of $8.4°$. The experiments of Section VII use it to quantify how a collagen cover distorts ME-HSI signals and how acquisitions can be designed to be robust. [TODO: one-sentence statement of the main experimental conclusion].

Code Availability

The source code, documentation, validation scripts and presets are available at https://github.com/Nard248/spectral-select.

References

[1] W. R. Zipfel, R. M. Williams, R. Christie, A. Y. Nikitin, B. T. Hyman, and W. W. Webb, "Live tissue intrinsic emission microscopy using multiphoton-excited native fluorescence and second harmonic generation," *Proceedings of the National Academy of Sciences*, vol. 100, no. 12, pp. 7075–7080, 2003.

[2] I. Georgakoudi and K. P. Quinn, "Optical Imaging Using Endogenous Contrast to Assess Metabolic State," *Annual Review of Biomedical Engineering*, vol. 14, no. 1, pp. 351–367, 2012.

[3] A. Croce and G. Bottiroli, "Autofluorescence spectroscopy and imaging: a tool for biomedical research and diagnosis," *European Journal of Histochemistry*, 2014.

[4] N. Ramanujam, "Fluorescence Spectroscopy of Neoplastic and Non-Neoplastic Tissues," *Neoplasia*, vol. 2, no. 1-2, pp. 89–117, 2000.

[5] M. Mercader, L. Swift, S. Sood, H. Asfour, M. Kay, and N. Sarvazyan, "Use of endogenous NADH fluorescence for real-time in situ visualization of epicardial radiofrequency ablation lesions and gaps," *American Journal of Physiology-Heart and Circulatory Physiology*, vol. 302, no. 10, pp. H2131–H2138, 2012.

[6] D. A. Gil, L. M. Swift, H. Asfour, N. Muselimyan, M. A. Mercader, and N. A. Sarvazyan, "Autofluorescence hyperspectral imaging of radiofrequency ablation lesions in porcine cardiac tissue," *Journal of Biophotonics*, vol. 10, no. 8, pp. 1008–1017, 2017.

[7] S. Guan, H. Asfour, N. Sarvazyan, and M. Loew, "Application of unsupervised learning to hyperspectral imaging of cardiac ablation lesions," *Journal of Medical Imaging*, vol. 5, no. 04, pp. 1–12, 2018.

[8] N. Chilingaryan, F. Villarruel, T. Soghomonyan, V. Yeghiazaryan, and N. Sarvazyan, "4D hyperspectral imaging for intraoperative tissue classification," in *Medical Imaging 2025: Clinical and Biomedical Imaging*, 2025, p. 35.

[9] N. Chilingaryan, A. Gasparyan, and N. Sarvazyan, "Multi-Excitation Hyperspectral Imaging Toward Improved Lichen Identification," in *2025 15th Workshop on Hyperspectral Imaging and Signal Processing: Evolution in Remote Sensing (WHISPERS)*, 2025, pp. 1–5.

[10] A. Hunanyan, N. Ter-Petrosyan, F. Villarruel, T. Soghomonyan, N. Sarvazyan, A. Butavyan, and V. Yeghiazaryan, "Deep Learning Classification of Ablated Atrial Tissue in Multi-Excitation Hyperspectral Images," in *2025 15th Workshop on Hyperspectral Imaging and Signal Processing: Evolution in Remote Sensing (WHISPERS)*, 2025, pp. 1–5.

[11] M. C. Skala, K. M. Riching, A. Gendron-Fitzpatrick, J. Eickhoff, K. W. Eliceiri, J. G. White, and N. Ramanujam, "In vivo multiphoton microscopy of NADH and FAD redox states, fluorescence lifetimes, and cellular morphology in precancerous epithelia," *Proceedings of the National Academy of Sciences*, vol. 104, no. 49, pp. 19 494–19 499, 2007.

[12] J. Wu, M. S. Feld, and R. P. Rava, "Analytical model for extracting intrinsic fluorescence in turbid media," *Applied Optics*, vol. 32, no. 19, p. 3585, 1993.

[13] M. G. Müller, I. Georgakoudi, Q. Zhang, J. Wu, and M. S. Feld, "Intrinsic fluorescence spectroscopy in turbid media: disentangling effects of scattering and absorption," *Applied Optics*, vol. 40, no. 25, p. 4633, 2001.

[14] L. Wang, S. L. Jacques, and L. Zheng, "MCML—Monte Carlo modeling of light transport in multi-layered tissues," *Computer Methods and Programs in Biomedicine*, vol. 47, no. 2, pp. 131–146, 1995.

[15] Q. Fang and D. A. Boas, "Monte Carlo Simulation of Photon Migration in 3D Turbid Media Accelerated by Graphics Processing Units," *Optics Express*, vol. 17, no. 22, p. 20178, 2009.

[16] H. Shen and G. Wang, "A tetrahedron-based inhomogeneous Monte Carlo optical simulator," *Physics in Medicine and Biology*, vol. 55, no. 4, pp. 947–962, 2010.

[17] A. A. Leino, A. Pulkkinen, and T. Tarvainen, "ValoMC: a Monte Carlo software and MATLAB toolbox for simulating light transport in biological tissue," *OSA Continuum*, vol. 2, no. 3, p. 957, 2019.

[18] A. A. Goodenough and S. D. Brown, "DIRSIG5: Next-Generation Remote Sensing Data and Image Simulation Framework," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 10, no. 11, pp. 4818–4833, 2017.

[19] N. Meloyan and N. Sarvazyan, "SpectralForge: A physics-based simulator for fluorescence hyperspectral imaging," in *Proc. 16th Workshop on Hyperspectral Imaging and Signal Processing: Evolution in Remote Sensing (WHISPERS)*, 2026, iEEE, accepted.

[20] L. C. Henyey and J. L. Greenstein, "Diffuse radiation in the Galaxy," *The Astrophysical Journal*, vol. 93, p. 70, 1941.

[21] S. L. Jacques, "Monte Carlo Modeling of Light Transport in Tissue (Steady State and Time of Flight)," in *Optical-Thermal Response of Laser-Irradiated Tissue*. Springer Netherlands, 2011, pp. 109–144.

[22] S. L. Jacques and B. W. Pogue, "Tutorial on diffuse light transport," *Journal of Biomedical Optics*, vol. 13, no. 04, pp. 1–19, 2008.

[23] E. Alerstam, T. Svensson, and S. Andersson-Engels, "Parallel computing with graphics processing units for high-speed Monte Carlo simulation of photon migration," *Journal of Biomedical Optics*, vol. 13, no. 06, pp. 1–3, 2008.

[24] D. A. Boas, J. P. Culver, J. J. Stott, and A. K. Dunn, "Three dimensional Monte Carlo code for photon migration through complex heterogeneous media including the adult human head," *Optics Express*, vol. 10, no. 3, p. 159, 2002.

[25] Q. Fang, "Mesh-based Monte Carlo method using fast ray-tracing in Plücker coordinates," *Biomedical Optics Express*, vol. 1, no. 1, p. 165, 2010.

[26] S. Yan and Q. Fang, "Hybrid mesh and voxel based Monte Carlo algorithm for accurate and efficient photon transport modeling in complex bio-tissues," *Biomedical Optics Express*, vol. 11, no. 11, p. 6262, 2020.

[27] A. J. Welch, C. Gardner, R. Richards-Kortum, E. Chan, G. Criswell, J. Pfefer, and S. Warren, "Propagation of fluorescent light," *Lasers in Surgery and Medicine*, vol. 21, no. 2, pp. 166–178, 1997.

[28] S. Jacques, "Monte Carlo Simulations Of Fluorescence In Turbid Media," in *Handbook of Biomedical Fluorescence*. CRC Press, 2003.

[29] J. Swartling, A. Pifferi, A. M. K. Enejder, and S. Andersson-Engels, "Accelerated Monte Carlo models to simulate fluorescence spectra from layered tissues," *Journal of the Optical Society of America A*, vol. 20, no. 4, p. 714, 2003.

[30] A. Liebert, H. Wabnitz, N. Zolek, and R. Macdonald, "Monte Carlo algorithm for efficient simulation of time-resolved fluorescence in layered turbid media," *Optics Express*, vol. 16, no. 17, p. 13188, 2008.

[31] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain randomization for transferring deep neural networks from simulation to the real world," in *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 23–30.

[32] R. Bro, "PARAFAC. Tutorial and applications," *Chemometrics and Intelligent Laboratory Systems*, vol. 38, no. 2, pp. 149–171, 1997.

[33] K. R. Murphy, C. A. Stedmon, D. Graeber, and R. Bro, "Fluorescence spectroscopy and multi-way techniques. PARAFAC," *Analytical Methods*, vol. 5, no. 23, p. 6557, 2013.

[34] T. Zimmermann, "Spectral Imaging and Linear Unmixing in Light Microscopy," in *Advances in Biochemical Engineering/Biotechnology*. Springer Berlin Heidelberg, 2005, pp. 245–265.

[35] N. Keshava and J. Mustard, "Spectral unmixing," *IEEE Signal Processing Magazine*, vol. 19, no. 1, pp. 44–57, 2002.

[36] J. M. Bioucas-Dias, A. Plaza, N. Dobigeon, M. Parente, Q. Du, P. Gader, and J. Chanussot, "Hyperspectral Unmixing Overview: Geometrical, Statistical, and Sparse Regression-Based Approaches," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 5, no. 2, pp. 354–379, 2012.

[37] D. C. Heinz and C.-I. Chang, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 39, no. 3, pp. 529–545, 2001.

[38] C. L. Lawson and R. J. Hanson, *Solving Least Squares Problems*. Society for Industrial and Applied Mathematics, 1995.

[39] D. D. Lee and H. S. Seung, "Learning the parts of objects by non-negative matrix factorization," *Nature*, vol. 401, no. 6755, pp. 788–791, 1999.

[40] J. Nascimento and J. Dias, "Vertex component analysis: a fast algorithm to unmix hyperspectral data," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 43, no. 4, pp. 898–910, 2005.

[41] H. R. Morris, C. C. Hoyt, and P. J. Treado, "Imaging Spectrometers for Fluorescence and Raman Microscopy: Acousto-Optic and Liquid Crystal Tunable Filters," *Applied Spectroscopy*, vol. 48, no. 7, pp. 857–866, 1994.

[42] G. Lu and B. Fei, "Medical hyperspectral imaging: a review," *Journal of Biomedical Optics*, vol. 19, no. 01, pp. 1–24, 2014.

[43] A. M. Valm, S. Cohen, W. R. Legant, J. Melunis, U. Hershberg, E. Wait, A. R. Cohen, M. W. Davidson, E. Betzig, and J. Lippincott-Schwartz, "Applying systems-level spectral imaging and analysis to reveal the organelle interactome," *Nature*, vol. 546, no. 7656, pp. 162–167, 2017.

[44] R. Richards-Kortum and E. Sevick-Muraca, "Quantitative optical spectroscopy for tissue diagnosis," *Annual Review of Physical Chemistry*, vol. 47, no. 1, pp. 555–606, 1996.

[45] I. Sobol', "Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates," *Mathematics and Computers in Simulation*, vol. 55, no. 1-3, pp. 271–280, 2001.

[46] A. Saltelli, P. Annoni, I. Azzini, F. Campolongo, M. Ratto, and S. Tarantola, "Variance based sensitivity analysis of model output. Design and estimator for the total sensitivity index," *Computer Physics Communications*, vol. 181, no. 2, pp. 259–270, 2010.

[47] M. J. Jansen, "Analysis of variance designs for model output," *Computer Physics Communications*, vol. 117, no. 1-2, pp. 35–43, 1999.

[48] M. D. Morris, "Factorial Sampling Plans for Preliminary Computational Experiments," *Technometrics*, vol. 33, no. 2, pp. 161–174, 1991.

[49] F. Campolongo, J. Cariboni, and A. Saltelli, "An effective screening design for sensitivity analysis of large models," *Environmental Modelling & Software*, vol. 22, no. 10, pp. 1509–1518, 2007.

[50] F. Pukelsheim, *Optimal Design of Experiments*. Society for Industrial and Applied Mathematics, 2006.

[51] G. L. Nemhauser, L. A. Wolsey, and M. L. Fisher, "An analysis of approximations for maximizing submodular set functions—I," *Mathematical Programming*, vol. 14, no. 1, pp. 265–294, 1978.

[52] C.-I. Chang, Q. Du, T.-L. Sun, and M. L. G. Althouse, "A joint band prioritization and band-decorrelation approach to band selection for hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 37, no. 6, pp. 2631–2641, 1999.

[53] W. Sun and Q. Du, "Hyperspectral Band Selection: A Review," *IEEE Geoscience and Remote Sensing Magazine*, vol. 7, no. 2, pp. 118–139, 2019.

[54] Y. Cai, X. Liu, and Z. Cai, "BS-Nets: An End-to-End Framework for Band Selection of Hyperspectral Image," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 58, no. 3, pp. 1969–1984, 2020.

[55] N. Meloyan and N. Sarvazyan, "Unsupervised deep learning for wavelength selection in multi-excitation fluorescence imaging," *Scientific Reports*, 2025, under revision.
[56] A. Rasheed, O. San, and T. Kvamsdal, "Digital Twin: Values, Challenges and Enablers From a Modeling Perspective," *IEEE Access*, vol. 8, pp. 21 980–22 012, 2020.
[57] E. Katsoulakis, Q. Wang, H. Wu, L. Shahriyari, R. Fletcher, J. Liu, L. Achenie, H. Liu, P. Jackson, Y. Xiao, T. Syeda-Mahmood, R. Tuli, and J. Deng, "Digital twins for health: a scoping review," *npj Digital Medicine*, vol. 7, no. 1, p. 77, 2024.
[58] R. L. Cook and K. E. Torrance, "A Reflectance Model for Computer Graphics," *ACM Transactions on Graphics*, vol. 1, no. 1, pp. 7–24, 1982.
[59] B. Smith, "Geometrical shadowing of a random rough surface," *IEEE Transactions on Antennas and Propagation*, vol. 15, no. 5, pp. 668–671, 1967.
[60] L. Song, E. Hennink, I. Young, and H. Tanke, "Photobleaching kinetics of fluorescein in quantitative fluorescence microscopy," *Biophysical Journal*, vol. 68, no. 6, pp. 2588–2600, 1995.
[61] C. Eggeling, J. Widengren, R. Rigler, and C. A. M. Seidel, "Photobleaching of Fluorescent Dyes under Conditions Used for Single-Molecule Detection: Evidence of Two-Step Photolysis," *Analytical Chemistry*, vol. 70, no. 13, pp. 2651–2659, 1998.
[62] E. Alerstam, S. Andersson-Engels, and T. Svensson, "White Monte Carlo for time-resolved photon migration," *Journal of Biomedical Optics*, vol. 13, no. 04, pp. 1–10, 2008.
[63] S. L. Jacques, "Optical properties of biological tissues: a review," *Physics in Medicine and Biology*, vol. 58, no. 11, pp. R37–R61, 2013.
[64] R. M. Pope and E. S. Fry, "Absorption spectrum (380–700 nm) of pure water II Integrating cavity measurements," *Applied Optics*, vol. 36, no. 33, p. 8710, 1997.
[65] G. M. Hale and M. R. Querry, "Optical Constants of Water in the 200-nm to 200-μm Wavelength Region," *Applied Optics*, vol. 12, no. 3, p. 555, 1973.
[66] R. L. van Veen, H. J. Sterenborg, A. Pifferi, A. Torricelli, E. Chikoidze, and R. Cubeddu, "Determination of visible near-IR absorption coefficients of mammalian fat using time- and spatially resolved diffuse reflectance and transmission spectroscopy," *Journal of Biomedical Optics*, vol. 10, no. 05, pp. 1–6, 2005.
[67] H. J. van Staveren, C. J. M. Moes, J. van Marie, S. A. Prahl, and M. J. C. van Gemert, "Light scattering in Intralipid-10% in the wavelength range of 400–1100 nm," *Applied Optics*, vol. 30, no. 31, p. 4507, 1991.
[68] R. Michels, F. Foschum, and A. Kienle, "Optical properties of fat emulsions," *Optics Express*, vol. 16, no. 8, p. 5907, 2008.
[69] E. Veach and L. J. Guibas, "Optimally combining sampling techniques for Monte Carlo rendering," in *Proceedings of the 22nd annual conference on Computer graphics and interactive techniques - SIGGRAPH '95*, 1995, pp. 419–428.
[70] J. R. Janesick, *Photon Transfer*. SPIE, 2007.
[71] G. Healey and R. Kondepudy, "Radiometric CCD camera calibration and noise estimation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 16, no. 3, pp. 267–276, 1994.
[72] D. J. Venzon and S. H. Moolgavkar, "A Method for Computing Profile-Likelihood-Based Confidence Intervals," *Applied Statistics*, vol. 37, no. 1, p. 87, 1988.
[73] G. H. Golub and V. Pereyra, "The Differentiation of Pseudo-Inverses and Nonlinear Least Squares Problems Whose Variables Separate," *SIAM Journal on Numerical Analysis*, vol. 10, no. 2, pp. 413–432, 1973.
[74] T. J. Farrell, M. S. Patterson, and B. Wilson, "A diffusion theory model of spatially resolved, steady-state diffuse reflectance for the noninvasive determination of tissue optical properties in vivo," *Medical Physics*, vol. 19, no. 4, pp. 879–888, 1992.
[75] M. S. Patterson, B. Chance, and B. C. Wilson, "Time resolved reflectance and transmittance for the noninvasive measurement of tissue optical properties," *Applied Optics*, vol. 28, no. 12, p. 2331, 1989.
[76] M. Kubista, R. Sjöback, S. Eriksson, and B. Albinsson, "Experimental correction for the inner-filter effect in fluorescence spectra," *The Analyst*, vol. 119, no. 3, pp. 417–419, 1994.